

COASTAL DYNAMICS

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- **Outline:**

1. Introduction
2. General equations
3. Surface Waves
 - 3.1. Surface gravity waves
 - 3.2. Inertia - gravity Waves
 - 3.3. Coastal waves
4. Tides
5. Internal Waves
6. Geostrophy and thermal wind
7. Bottom and surface boundary layers
8. Coastal circulation and responses to meteorological forcing
9. Frontal dynamics
10. Estuary plumes and regimes
11. Climate change

Presentations and material will be available at :

jgula.fr/Coastal/

3. Surface WAVES

3.1.Introduction

3.1.1.General properties of Wave

3.1.2. Different types of ocean waves

3.2.Surface Gravity Waves

3.2.1.Long waves

3.2.2.Short waves

3.3.Inertia-gravity Waves

3.4.Coastal waves

3.2 Surface Gravity Waves

- **See notes:**
 - *Solution for deep-water waves*

Review of the results of our analysis of surface gravity waves on an infinitely deep ocean

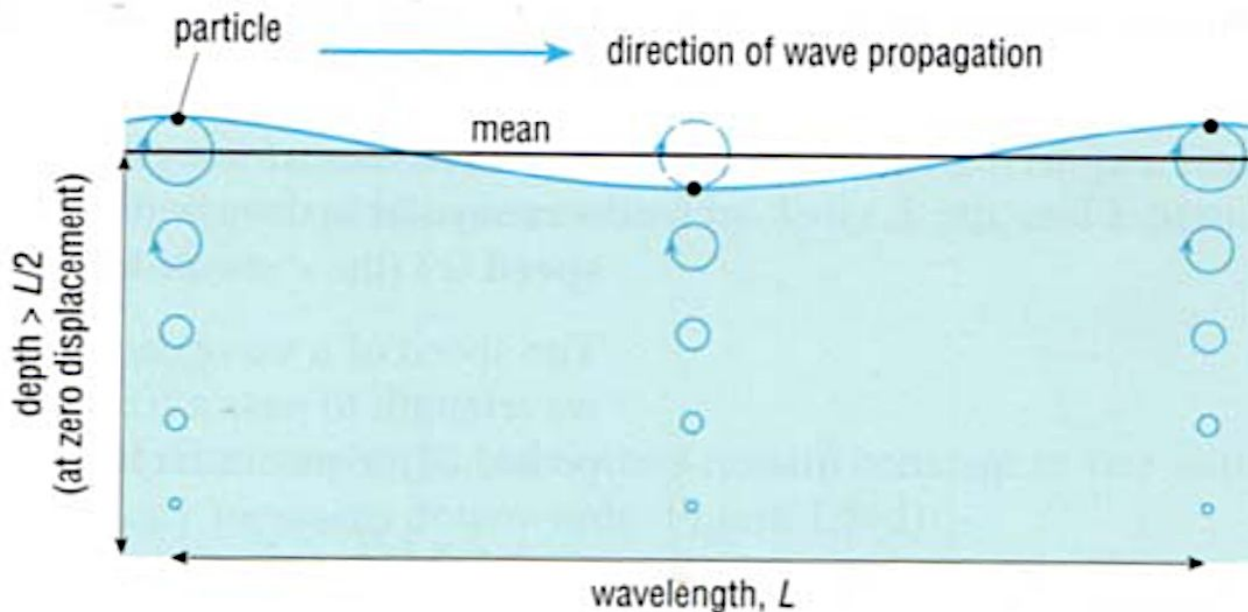
Our analysis shows that:

1. The particles follow circular paths, the amplitudes of which decrease exponentially with depth

$$x' = -Ae^{k\bar{z}} \sin(k\bar{x} - \omega t)$$

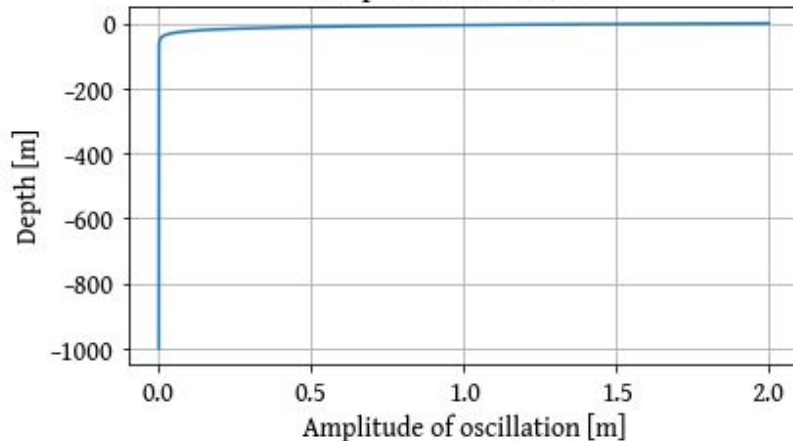
$$z' = Ae^{k\bar{z}} \cos(k\bar{x} - \omega t)$$

The particles execute circular motions: they do not move with the propagating wave (this is true to leading order: the Stokes drift that we talked about previously is a 2nd order effect)

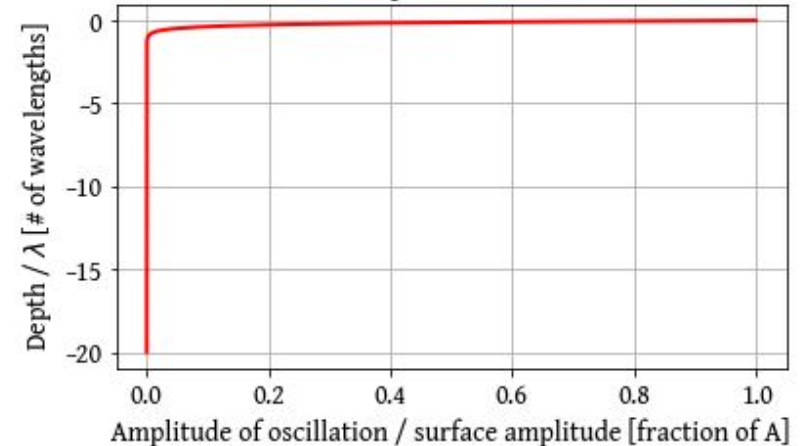


Review of the results of our analysis of surface gravity waves on an infinitely deep ocean

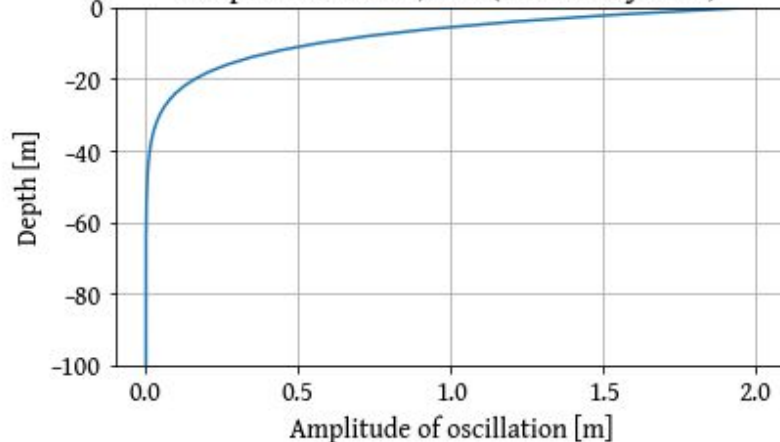
Amplitude of particle paths for an infinitely deep ocean: $\lambda=50$, $A=2$



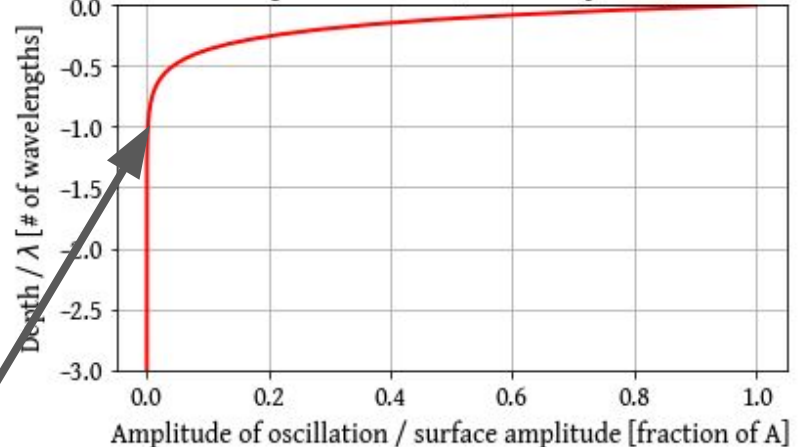
Amplitude of particle paths for an infinitely deep ocean: $A=2$



Amplitude of particle paths for an infinitely deep ocean: $\lambda=50$, $A=2$ (zoom on y-axis)



Amplitude of particle paths for an infinitely deep ocean: $A=2$ (zoom on y-axis)



e^{kz} dependence of the amplitude implies that almost all of the wave energy is contained within 0.5 - 1 wavelengths of the surface

Review of the results of our analysis of surface gravity waves on an infinitely deep ocean

2. The dispersion relationship is described by: $\omega = \sqrt{gk}$

- The phase speed is: $c_p = \frac{\omega}{k} = \sqrt{\frac{g}{k}}$ The group speed is: $c_g = \frac{\partial \omega}{\partial k} = \frac{1}{2} \sqrt{\frac{g}{k}}$
- The group speed is half of the phase speed

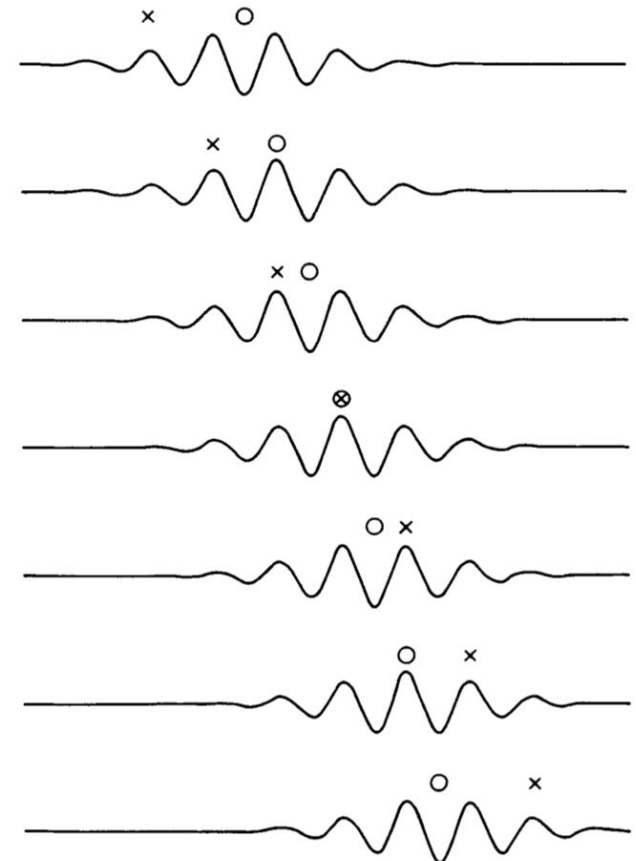
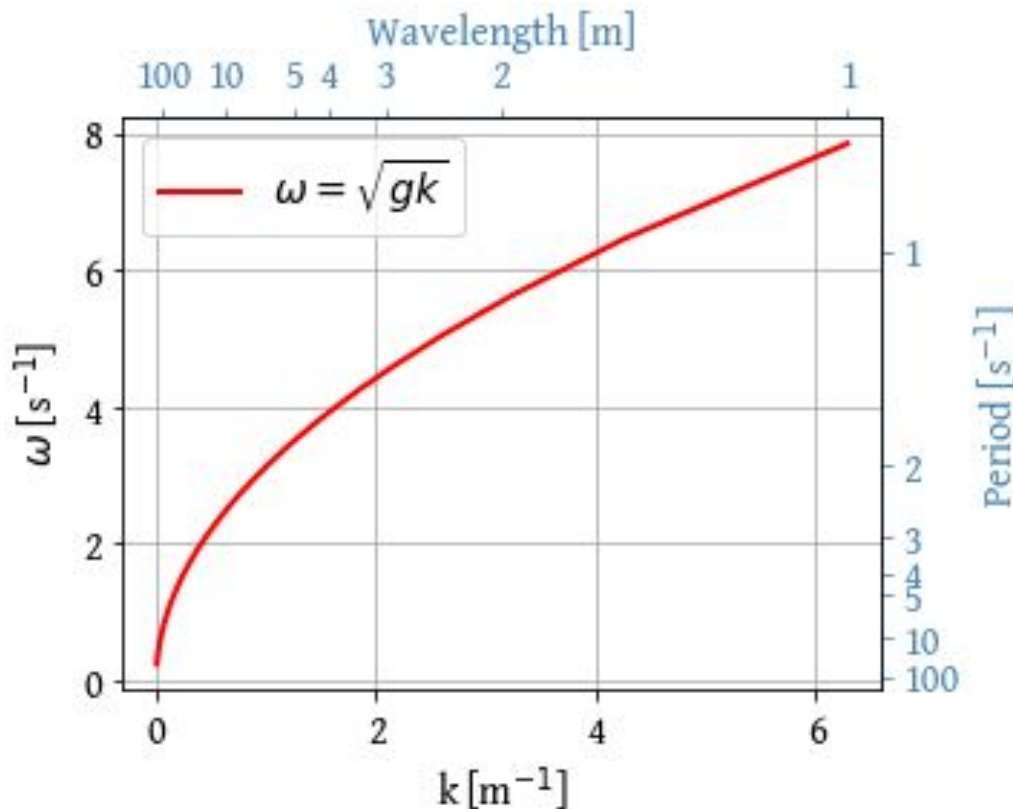


Fig. 12.3 A travelling wave group. The diagrams show the group at times one period apart. In this example the dispersion is 'normal', with the group velocity equal to half the phase velocity. A specimen wave crest (marked $\times$) enters the group at the trailing edge; its amplitude increases to a maximum and then decreases again as it passes forwards through the group, to disappear again at the leading edge of the group. The position of maximum disturbance in the group is marked $\circ$ in each picture.

Review of the results of our analysis of surface gravity waves on an infinitely deep ocean

What does the dispersion relationship tell us practically?

We have seen that:

- surface gravity waves are dispersive
- long waves (lower k) travel faster than short waves (higher k)

Practically, this means that:

- an initial disturbance will spread out over time
- a certain time after a disturbance, the longest wavelengths will appear furthest away from the centre of the disturbance, and shortest wavelengths will appear nearest the centre



Review of the results of our analysis of surface gravity waves on an infinitely deep ocean



Longest waves propagate fastest

Wave crests get closer together over time

3.2.2 Short waves

Wind waves versus Swells

Feature	Wind Waves	Swells
Origin	Generated by local winds	Formed from decayed wind waves traveling away from storms
Wavelength	Shorter (meters to ~100 m)	Longer (100–500 m)
Period	Short (1–10 s)	Long (10–30 s)
Appearance	Steep, irregular, choppy	Smooth, uniform, regular
Direction	Aligned with local wind	Can travel across ocean basins, independent of local wind
Energy	Confined near wind source	Can transport energy thousands of km
Persistence	Disappear quickly when wind stops	Can last for days to weeks

3.2.2 Short waves

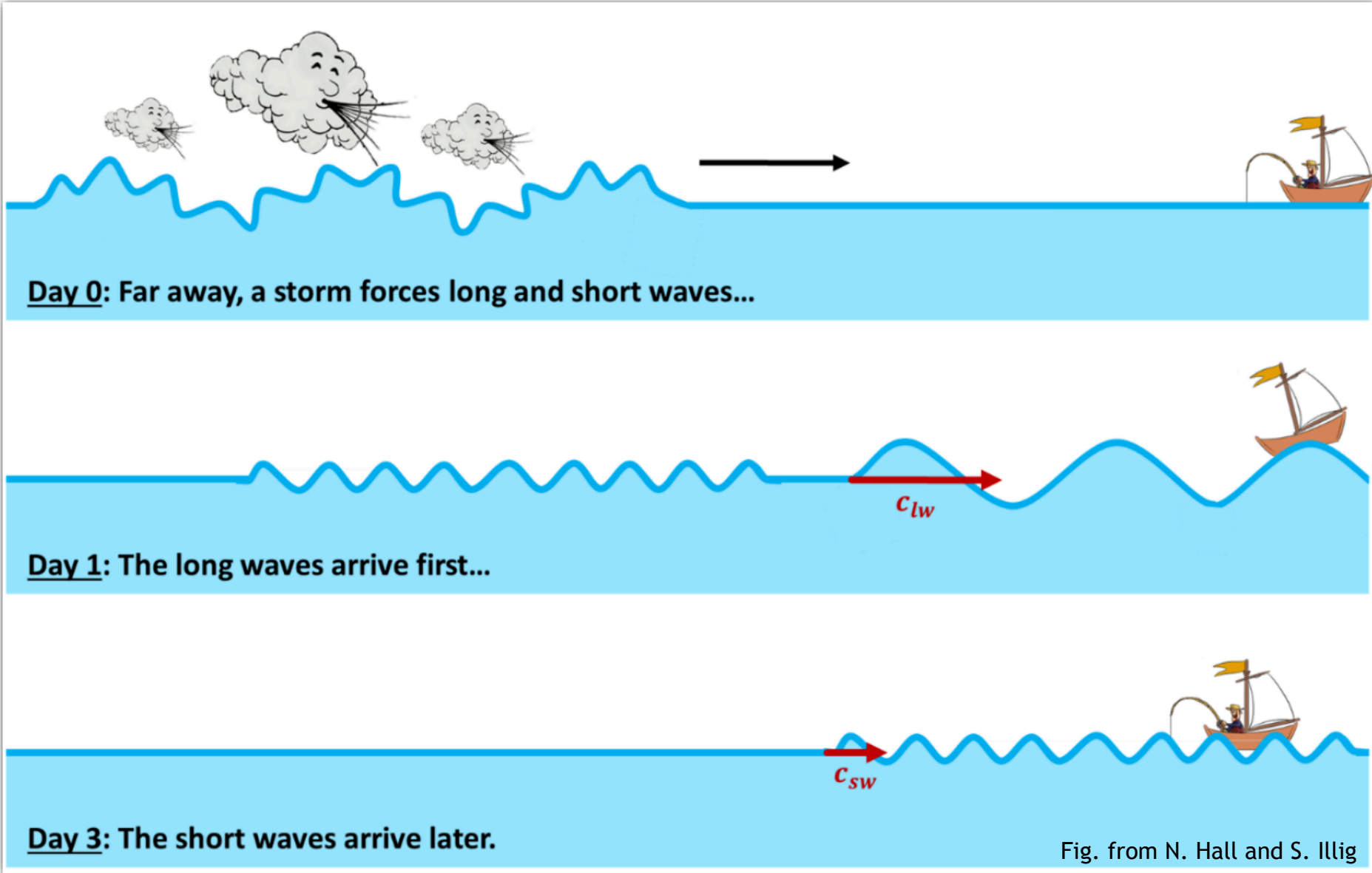


Fig. from N. Hall and S. Illig

3.2.2 Short waves

Munk's experiment in 1963: **"Waves Across the Pacific"**

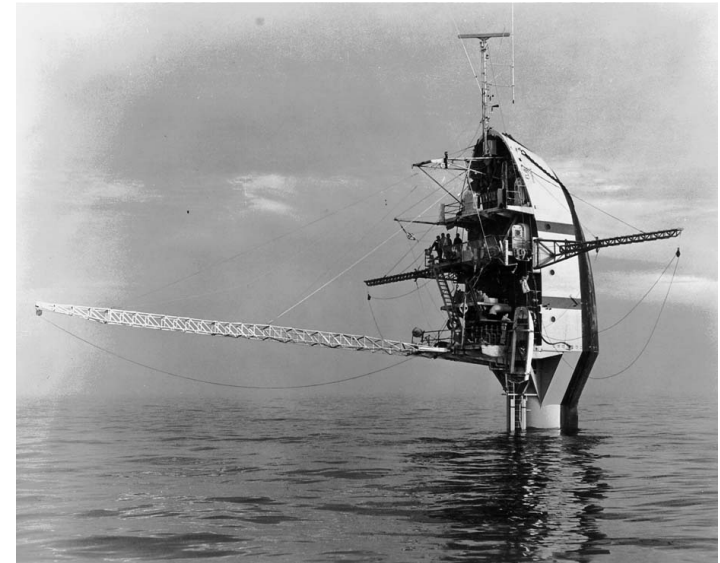
His equations said that the swells hitting beaches in Mexico began some 9,000 miles away — somewhere in the southern reaches of the Indian Ocean, near Antarctica.

"Could it be?" he wrote in an autobiographical sketch. Could a storm halfway across the world produce a patch of moving water that traveled from near the South Pole, up past Australia, then past New Zealand, then across the vast expanse of the Pacific, arriving still intact — at a beach off Mexico?

He decided to find out for himself.

That is why, in 1957, Walter Munk designed a global, real-life wave-watching experiment.

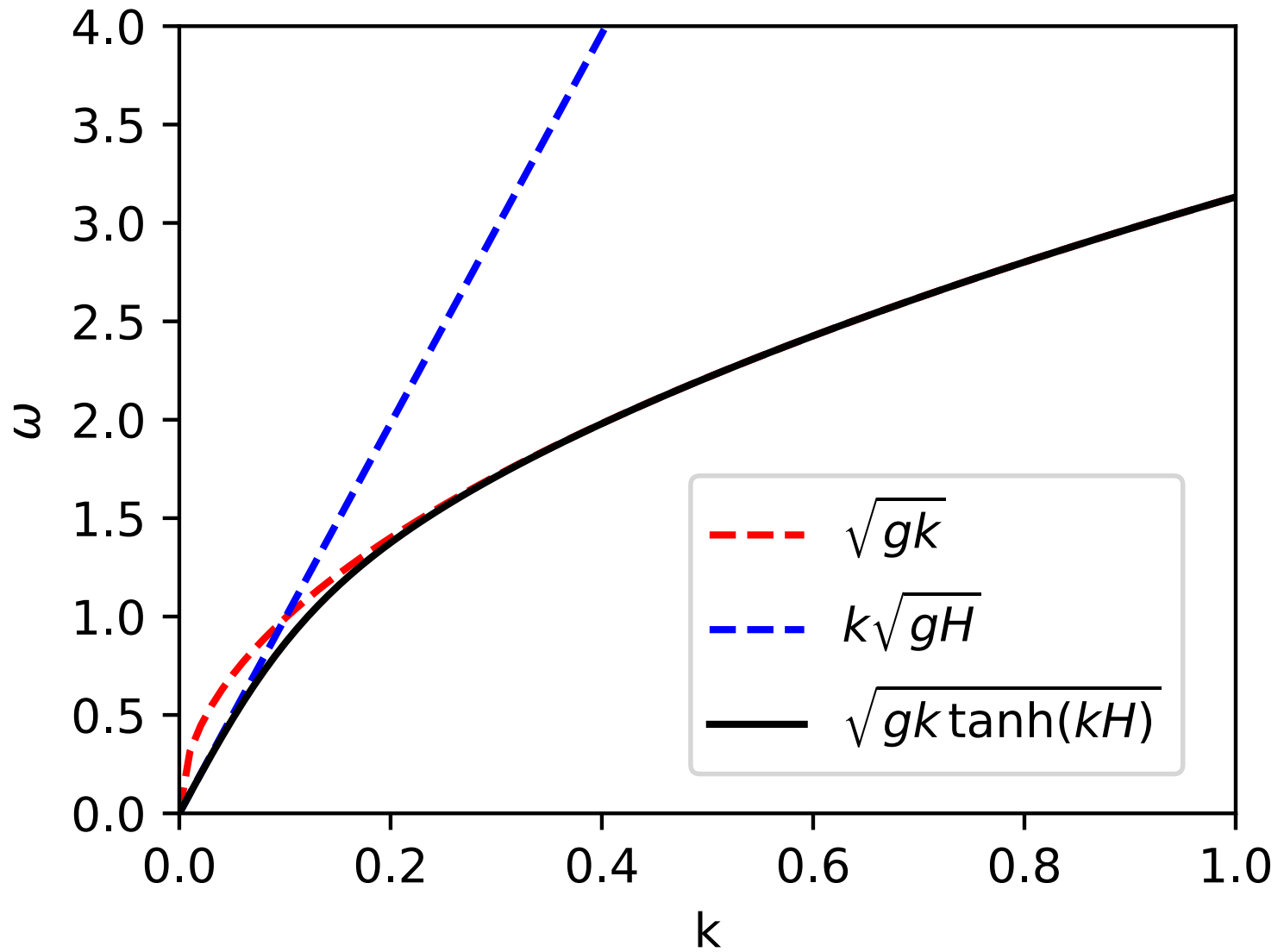
<https://www.youtube.com/watch?v=MX5cKoOm6Pk&pbjreload=10>



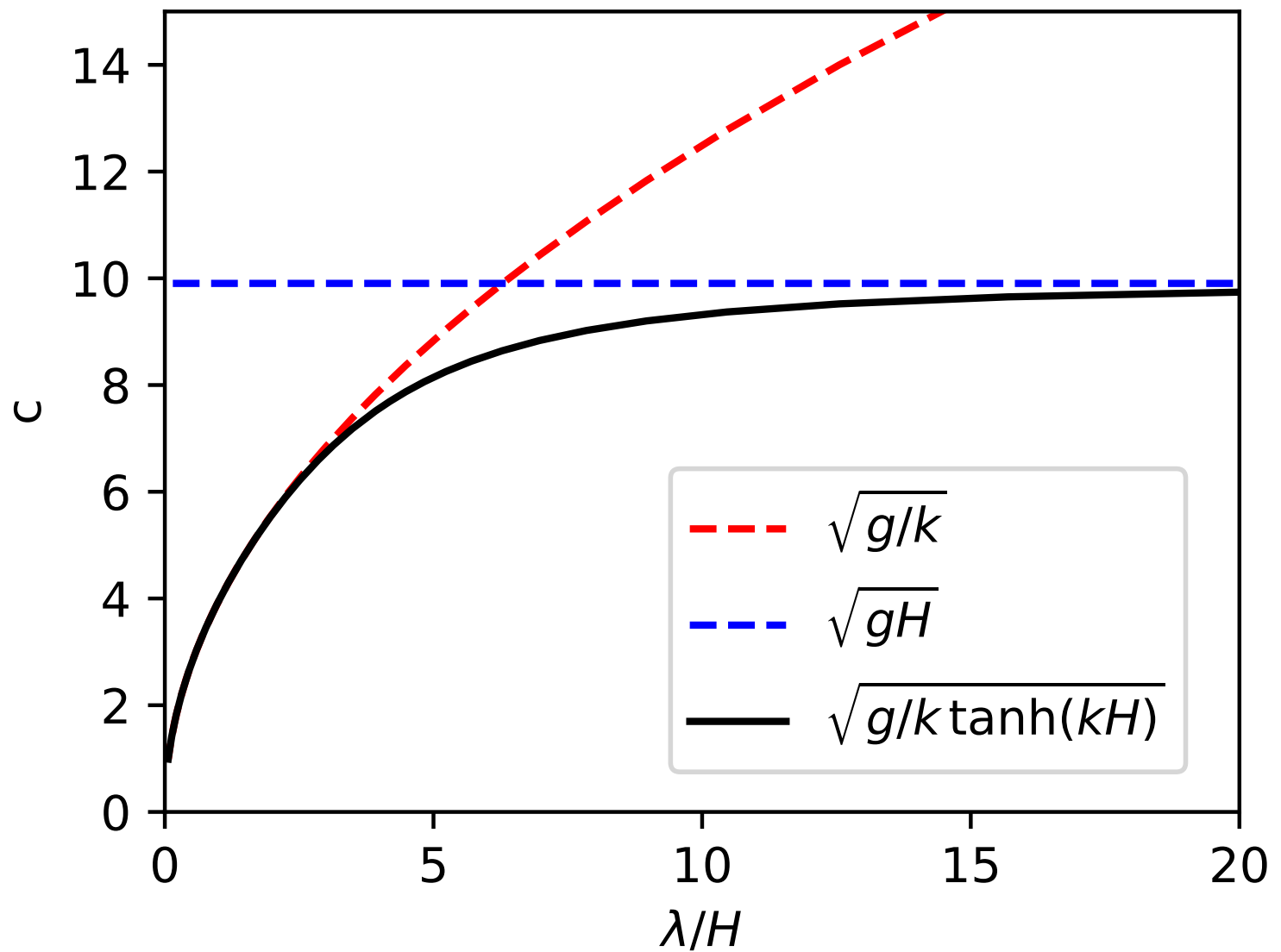
3.2.3 The general case

- **See notes:**
 - *Solution for the general case*

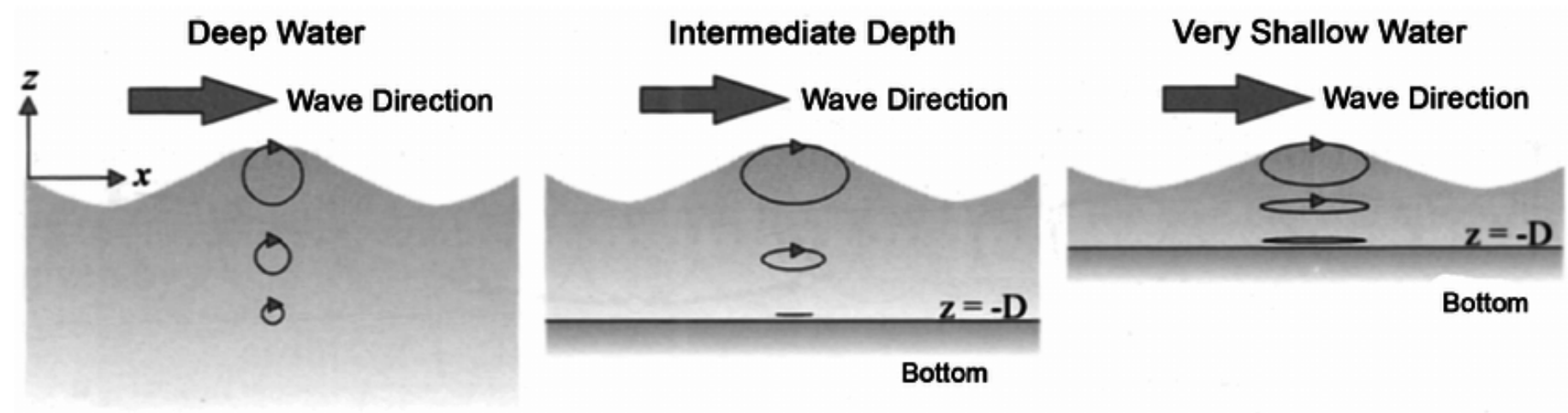
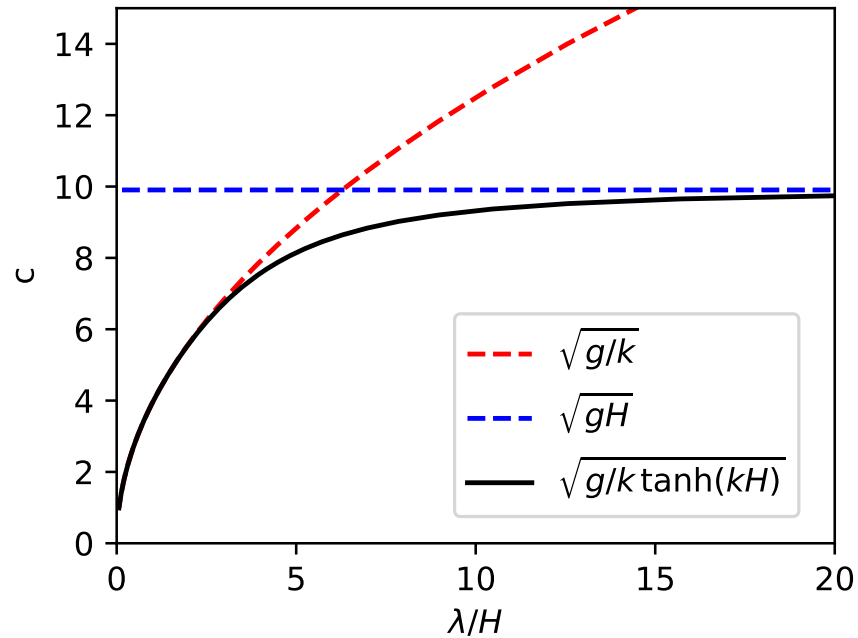
3.2.3 The general case



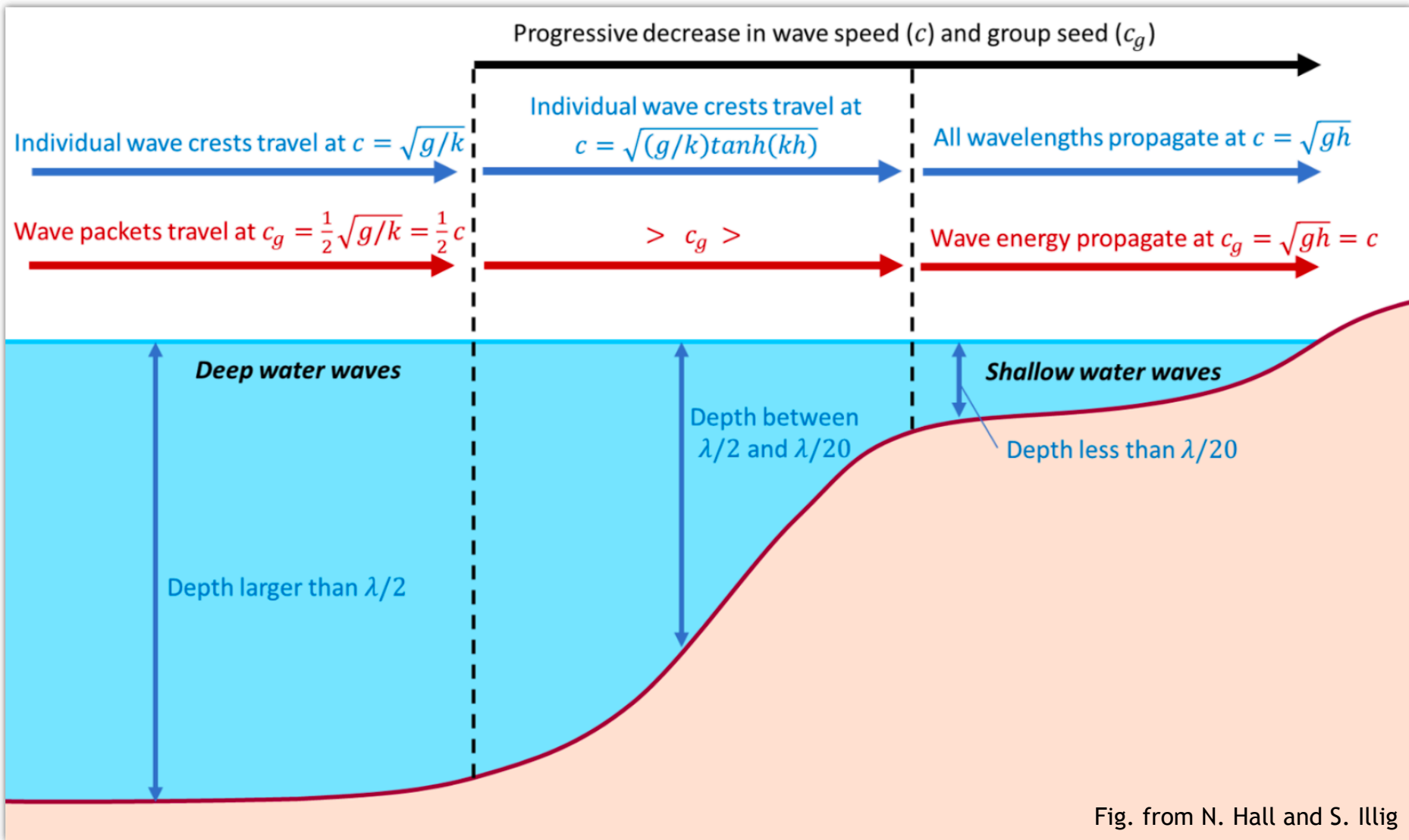
3.2.3 The general case



3.2.3 The general case

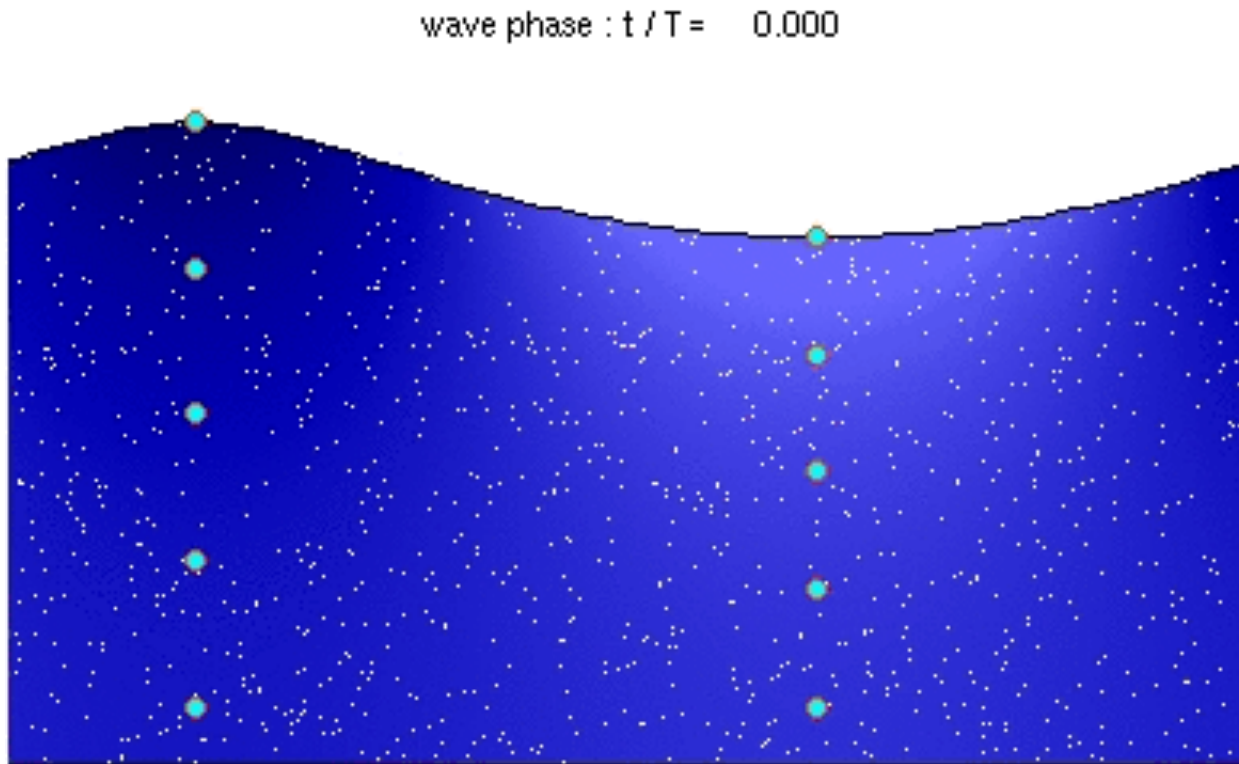


3.2.3 The general case



Waves change properties approaching the coast

3.2.3 The general case



A **nonlinear wave** also carries energy and can lead to a net transport of material
(for example, Stokes drift at the ocean surface)

Animation from https://en.wikipedia.org/wiki/Wind_wave

3.2.3 The general case

Stokes drift

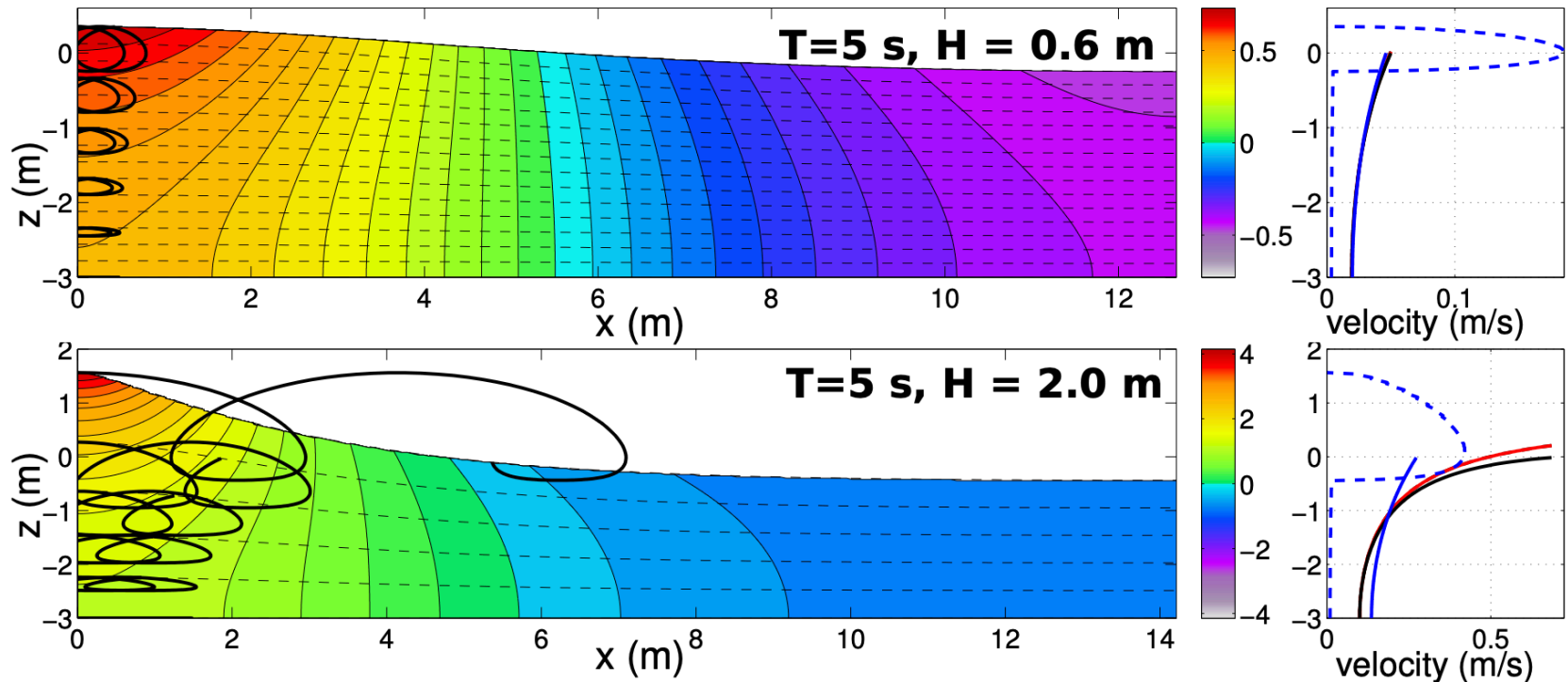


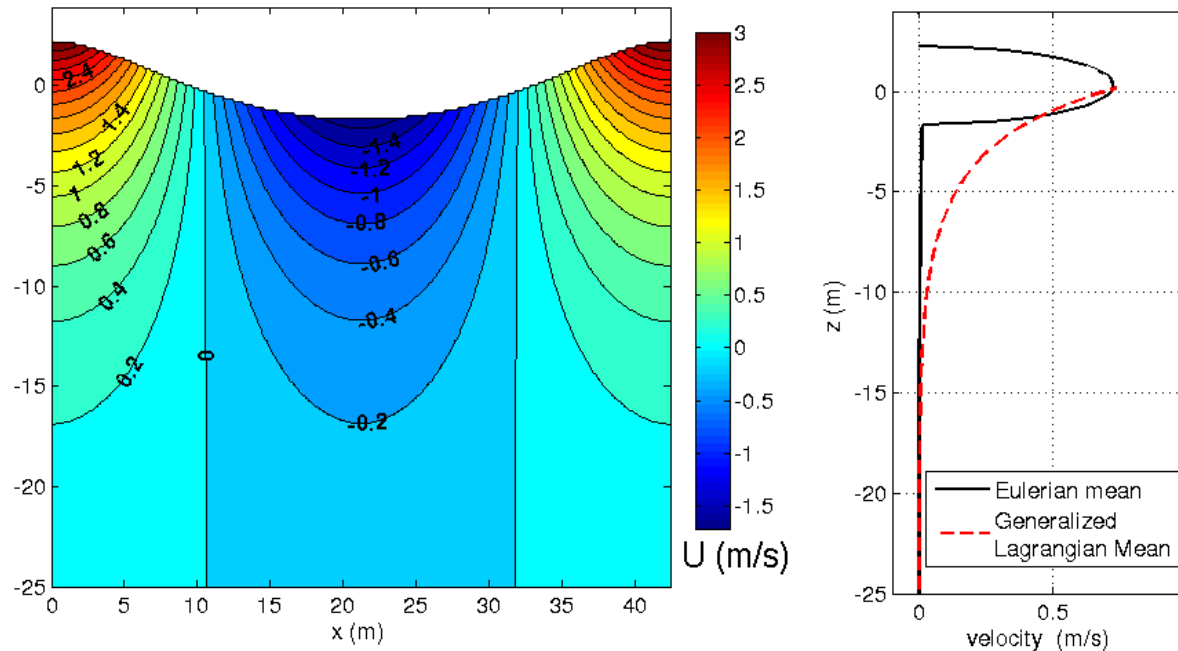
Figure 9.1: Left: Horizontal velocity field at $t = 0$ and particle trajectories integrated over 2 Eulerian periods. Solid lines are isotachs and dashed lines are streamlines in the frame of reference moving with the wave phase speed. Right: vertical profile of Eulerian mean water velocity (dashed, velocity is set to zero in the air for computing the average), Generalized Lagrangian mean (red), Lagrangian mean (black) and Lagrangian mean from the linear approximation (blue). Both top and bottom panels are computed with streamfunction theory to 80th order (Dalrymple, 1974). Non-linear terms are only significant in the bottom case.

3.2.3 The general case

Stokes drift

For a long wave, with surface elevation: $\eta = a \cos(kx - \omega t)$

The velocity potential is $\phi = \frac{\omega}{k} a e^{kz} \sin(kx - \omega t)$



The Stokes drift velocity (Lagrangian minus Eulerian mean) is: $\bar{u}_s = \omega k a^2 e^{2kz}$

Wave energy flux analysis summary

- Wave energy density is “equipartitioned” : the kinetic and potential energy density contribute equally to the total energy density, and are proportional to the wave amplitude

$$E_k = \frac{1}{4}\rho g A^2 \quad E_p = \frac{1}{4}\rho g A^2 \quad E = E_k + E_p = \frac{1}{2}\rho g A^2$$

- Wave energy is transported at the group speed
- When wave energy is conserved, the changes in wave energy density over time and the flux are linked by the relationship:

$$\frac{\partial E}{\partial t} + \nabla \cdot (\vec{c}_g E) = 0$$

If the wind does work to generate new waves, or friction acts to destroy waves, then energy is not conserved and the above equation needs to be modified.

(bonus) Capillary waves

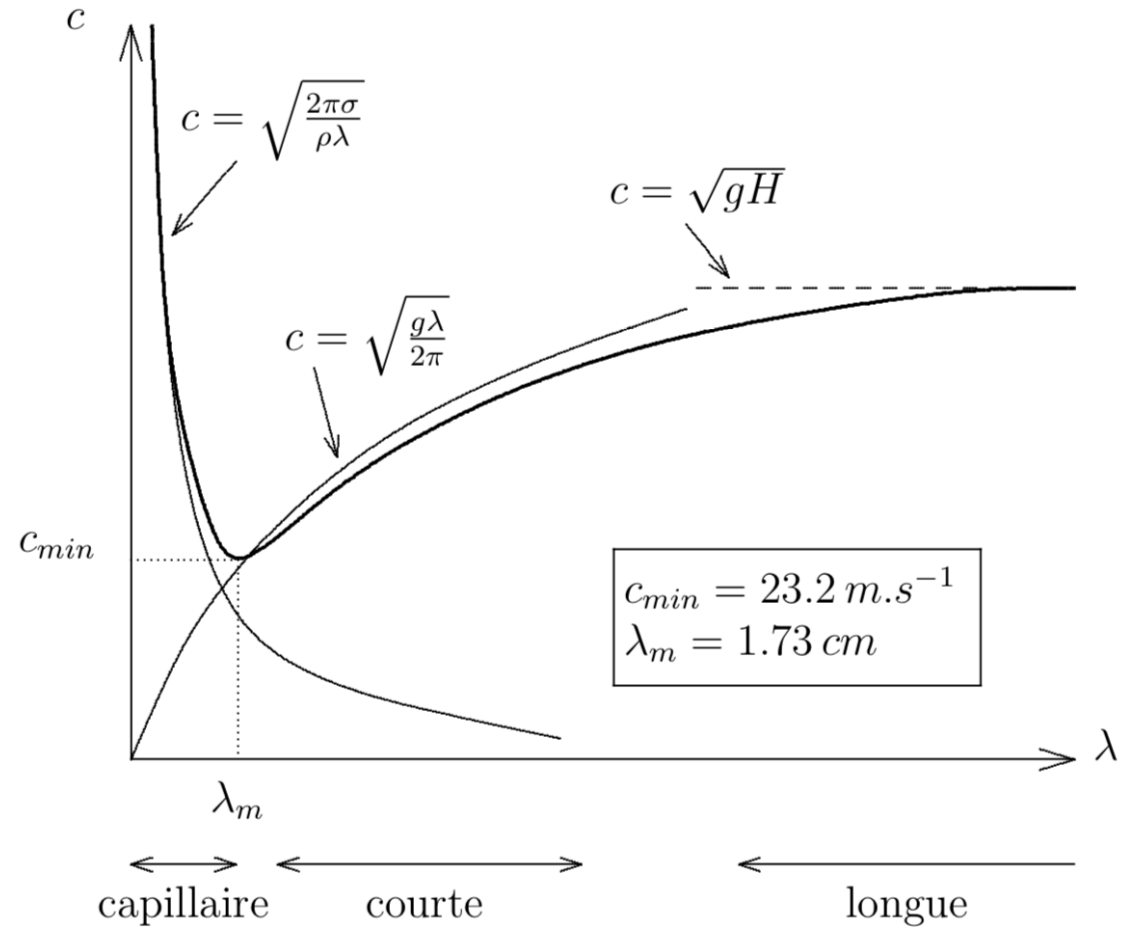


(bonus) Capillary waves



(bonus) Capillary waves

With surface tension effects:



3. Surface WAVES

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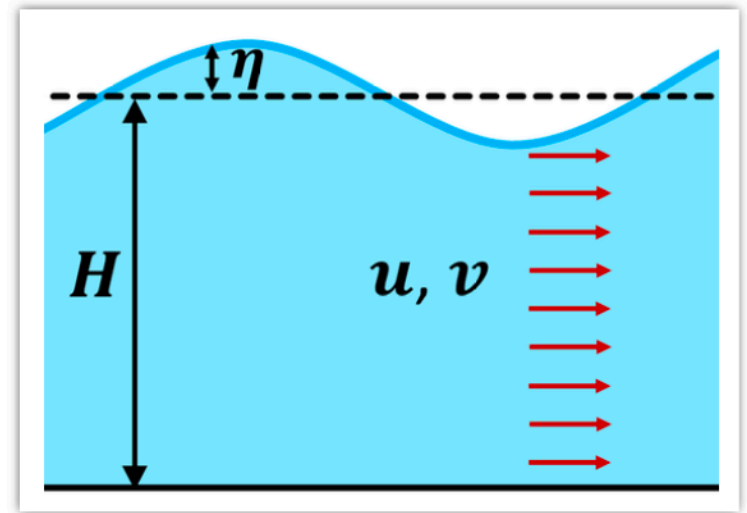
3.3.Inertia-gravity Waves

3.4.Coastal waves

3.3 Inertia-gravity Waves

Let's go back to the linear shallow water equations, but keep the rotation:

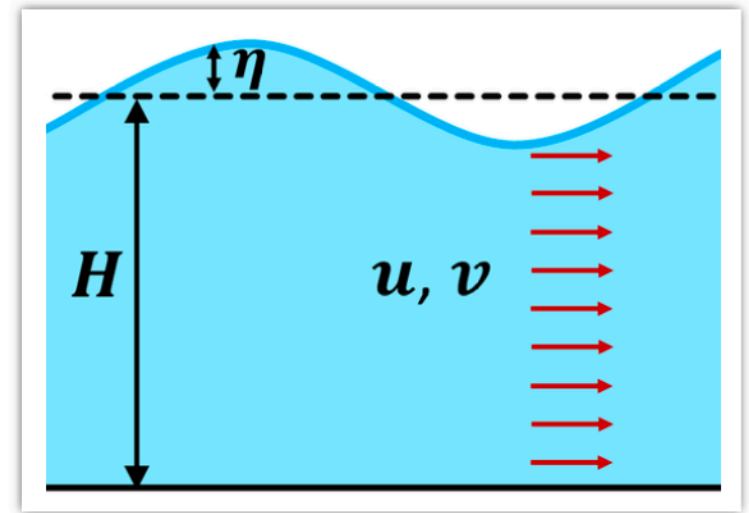
$$\begin{aligned}\frac{\partial u}{\partial t} - f v &= -g \frac{\partial \eta}{\partial x} \\ \frac{\partial v}{\partial t} + f u &= -g \frac{\partial \eta}{\partial y} \\ \frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) &= 0\end{aligned}$$



3.3.1 Inertial Oscillations

keep only the rotation:

$$\begin{aligned}\frac{\partial u}{\partial t} - fv &= -g \cancel{\frac{\partial \eta}{\partial x}} \\ \frac{\partial v}{\partial t} + fu &= -g \cancel{\frac{\partial \eta}{\partial y}} \\ \cancel{\frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)} &= 0\end{aligned}$$

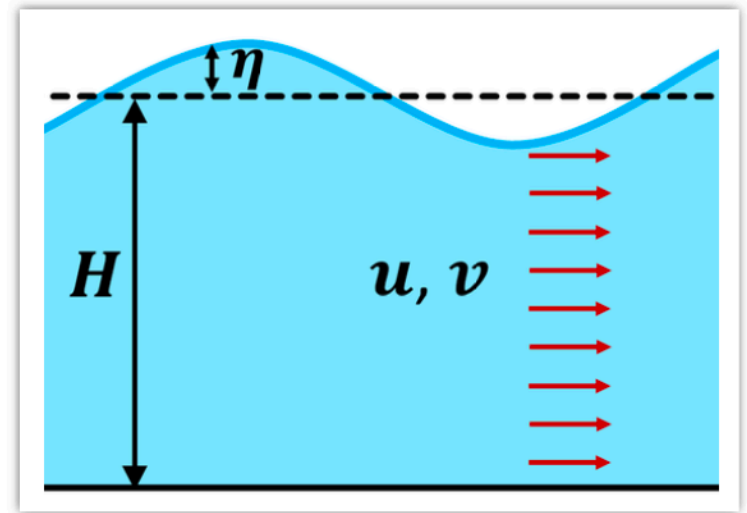


- **See notes:**
 - *Solution for inertial oscillations*

3.3 Inertia-gravity Waves

Let's go back to the linear shallow water equations, but keep the rotation:

$$\begin{aligned}\frac{\partial u}{\partial t} - fv &= -g \frac{\partial \eta}{\partial x} \\ \frac{\partial v}{\partial t} + fu &= -g \frac{\partial \eta}{\partial y} \\ \frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) &= 0\end{aligned}$$



- **See notes:**
 - *Solution for inertia-gravity waves*

3.3 Inertia-gravity Waves

Linear shallow water equations on an f-plane

$$u_t - fv + g\eta_x = 0 \quad (1)$$

$$v_t + fu + g\eta_y = 0 \quad (2)$$

$$\eta_t + H(u_x + v_y) = 0 \quad (3)$$

vorticity equation: $\frac{\partial}{\partial x}(2) - \frac{\partial}{\partial y}(1) \rightarrow \frac{\partial}{\partial t}(v_x - u_y) + f(u_x + v_y) = 0 \quad (V)$

divergence equation: $\frac{\partial}{\partial x}(1) + \frac{\partial}{\partial y}(2) \rightarrow \frac{\partial}{\partial t}(u_x + v_y) - f(v_x - u_y) + g\nabla^2\eta = 0 \quad (D)$

substitute (V) into (3) $\eta_t - \frac{H}{f} \frac{\partial}{\partial t}(v_x - u_y) = 0$

3.3 Inertia-gravity Waves

substitute (D) into $\frac{\partial}{\partial t}$ (3)

$$\eta_{tt} + fH(v_x - u_y) - gH\nabla^2\eta = 0 \quad \frac{\partial}{\partial t} \rightarrow$$

$$\eta_{ttt} + fH\frac{\partial}{\partial t}(v_x - u_y) - gH\nabla^2\eta_t = 0$$

substitute from $\eta_t - \frac{H}{f}\frac{\partial}{\partial t}(v_x - u_y) = 0$ gives $\eta_{ttt} + f^2\eta_t - gH\nabla^2\eta_t = 0$

With appropriate initial condition at $t = 0$, the departure from geostrophic disequilibrium follows:

$$\eta_{tt} - gH\nabla^2\eta + f^2\eta = 0 \quad \text{substitute solution} \quad \eta = \tilde{\eta}e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$$

leads to dispersion relation

$$\omega = \pm\sqrt{f^2 + gHk^2}$$

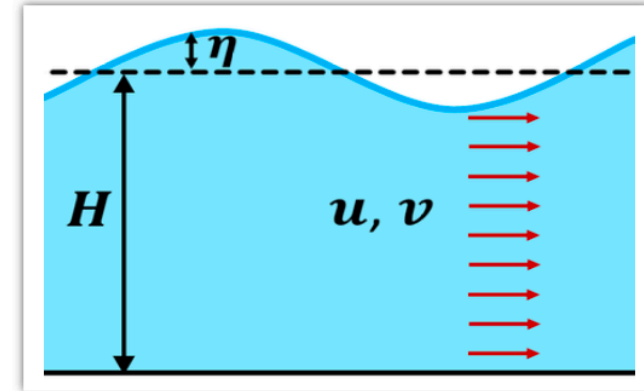
3.3 Inertia-gravity Waves

⇒ flat bottom, f -plane, linear perturbations u, v, η

$$\frac{\partial u}{\partial t} - fv = -g \frac{\partial \eta}{\partial x}$$

$$\frac{\partial v}{\partial t} + fu = -g \frac{\partial \eta}{\partial y}$$

$$\frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0$$



Wave solution: $(u, v, \eta) = (\tilde{u}, \tilde{v}, \tilde{\eta})e^{i(lx+my-\omega t)}$

With: $\frac{\partial}{\partial x} \rightarrow il \times$ $\frac{\partial}{\partial y} \rightarrow im \times$ $\frac{\partial}{\partial t} \rightarrow -i\omega \times$

⇒ substitute wave solution: differential equations become linear algebraic equations

$$-i\omega\tilde{u} - f\tilde{v} = -igl\tilde{\eta}$$

$$-i\omega\tilde{v} + f\tilde{u} = -igm\tilde{\eta}$$

$$-i\omega\tilde{\eta} + H(il\tilde{u} + im\tilde{v}) = 0$$

The unknowns are the wave amplitudes $\tilde{u}, \tilde{v}, \tilde{\eta}$

The parameters are the wave properties l, m, ω and the geophysical constants f, g, H

3.3 Inertia-gravity Waves

Alternative derivation:

We need to solve algebraic system

$$\begin{pmatrix} -i\omega & -f & igl \\ f & -i\omega & igm \\ ilH & imH & -i\omega \end{pmatrix} \begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{\eta} \end{pmatrix} = 0$$

⇒ *trivial solution* $\tilde{u} = \tilde{v} = \tilde{\eta} = 0$ (no flow)

⇒ *The condition for having non-trivial solutions is that the determinant of the matrix is zero.*

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \cdot \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \cdot \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \cdot \det \begin{bmatrix} d & e \\ g & h \end{bmatrix}$$

This leads to $\omega [\omega^2 - f^2 - gH(l^2 + m^2)] = 0$

this is a more complicated dispersion relation !

three solutions: $\omega = 0$ steady geostrophic flow

$$\omega = \pm \sqrt{f^2 + gHk^2} \text{ inertia-gravity waves}$$

3.3 Inertia-gravity Waves

short wave limit: $k^2 \gg f^2/gH$

i.e. the wavelength $\lambda^2 \ll 2\pi L_d^2$
(L_d is the radius of deformation)

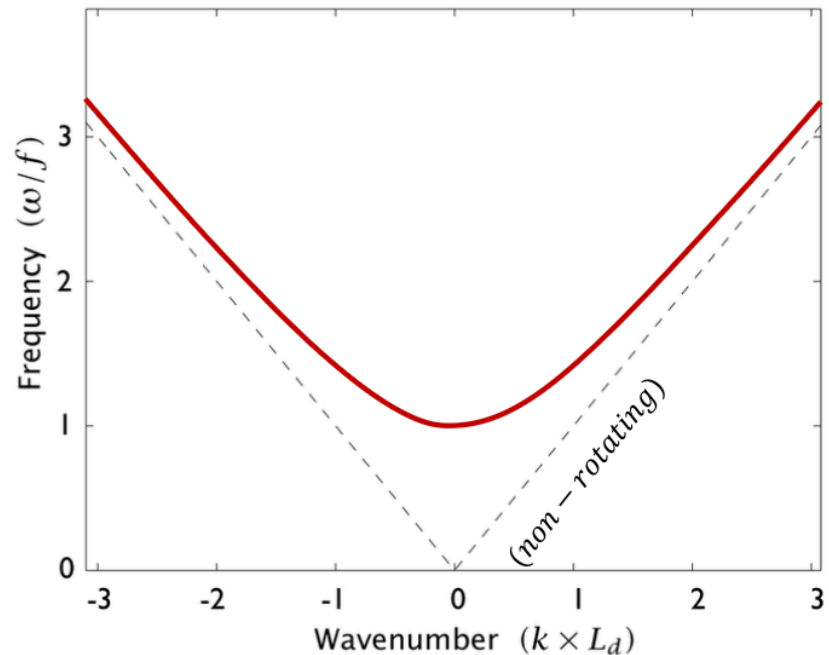
waves behave as in non-rotating case (provided shallow water condition is not violated)

long wave limit: $k^2 \ll f^2/gH$

dispersion relation reduced to

$$\omega = f$$

free inertial oscillations



3. Surface WAVES

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3.4.Coastal waves

Kelvin waves

What happens if you have a gravity wave along a solid boundary?

William Thompson (Lord Kelvin) was the first to wonder what happens if one imposes $u = 0$ (normal to the boundary) everywhere.

Starting from the linear SW equations, check what happens if $u = 0$:

$$\begin{aligned}\frac{\partial u}{\partial t} - fv &= -g \frac{\partial \eta}{\partial x} \\ \frac{\partial v}{\partial t} + fu &= -g \frac{\partial \eta}{\partial y} \\ \frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) &= 0\end{aligned}$$

3.4 Coastal waves

Kelvin waves

$$\cancel{\frac{\partial v}{\partial t}} - f v = -g \frac{\partial \eta}{\partial x} \quad \text{Geostrophic balance}$$

$$\cancel{\frac{\partial v}{\partial t}} + f u = -g \frac{\partial \eta}{\partial y}$$

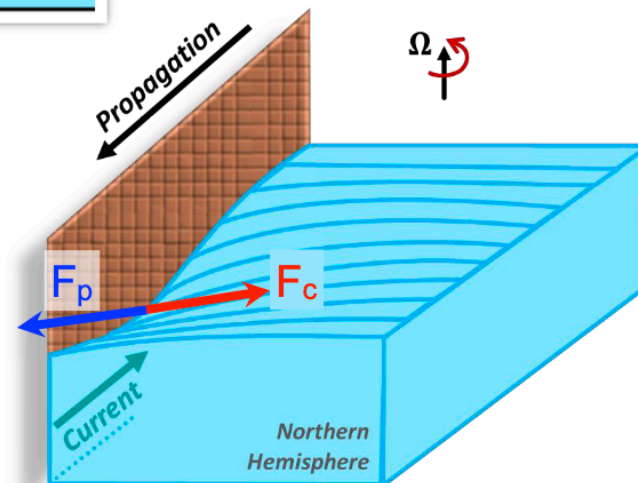
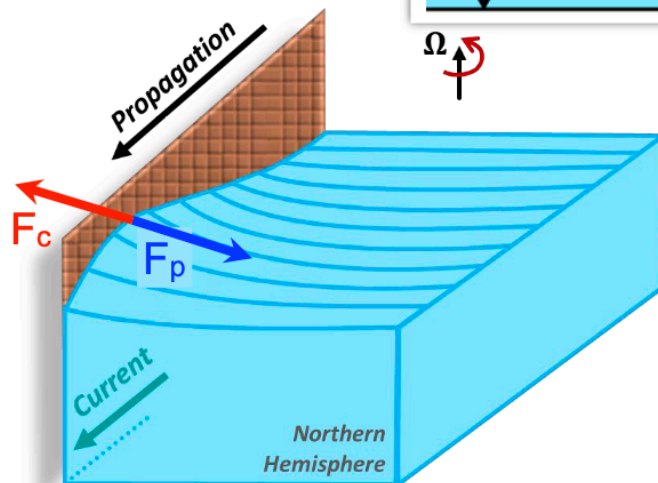
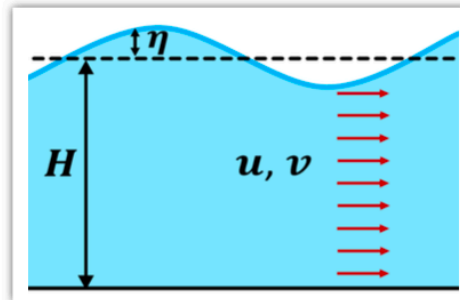
$$\cancel{\frac{\partial \eta}{\partial t}} + H \left(\cancel{\frac{\partial u}{\partial x}} + \frac{\partial v}{\partial y} \right) = 0$$

non-dispersive waves

In the x direction we have geostrophic balance, with pressure and Coriolis forces alternating in direction as crests and troughs propagate meridionally.

In the y direction we have non-dispersive gravity waves with a fixed phase speed independent of horizontal scale

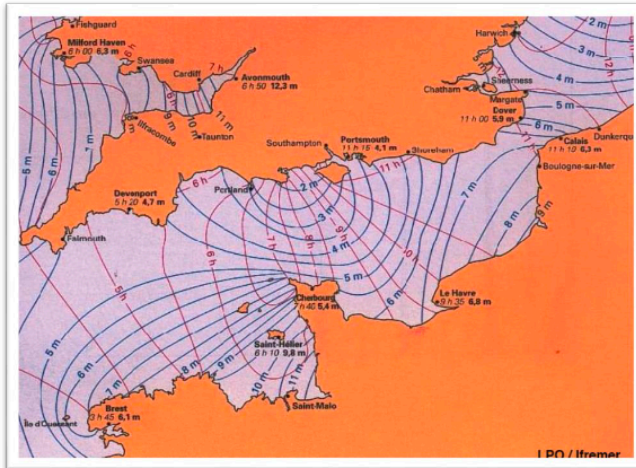
$$|c| = \sqrt{gH}$$



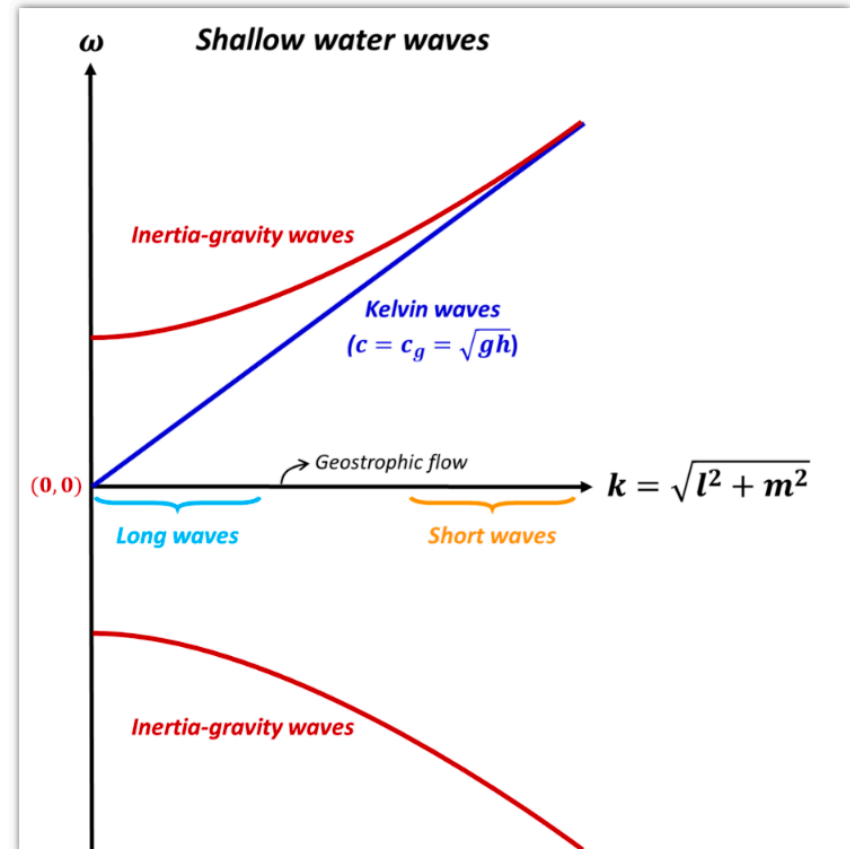
3.4 Coastal waves

Kelvin waves

Since the only admissible solution is V_1 , we conclude that for a system bounded on the west (x positive) the wave propagates in the negative y direction, i.e. to the south (in the northern hemisphere). If x is negative this reverses. Kelvin waves thus propagate south on the western boundary and north on the eastern boundary. They circuit ocean basins in an anticlockwise (cyclonic) direction. In the southern hemisphere the direction is clockwise (but still cyclonic).



Tides are higher on the French side because the signal propagates in from the west

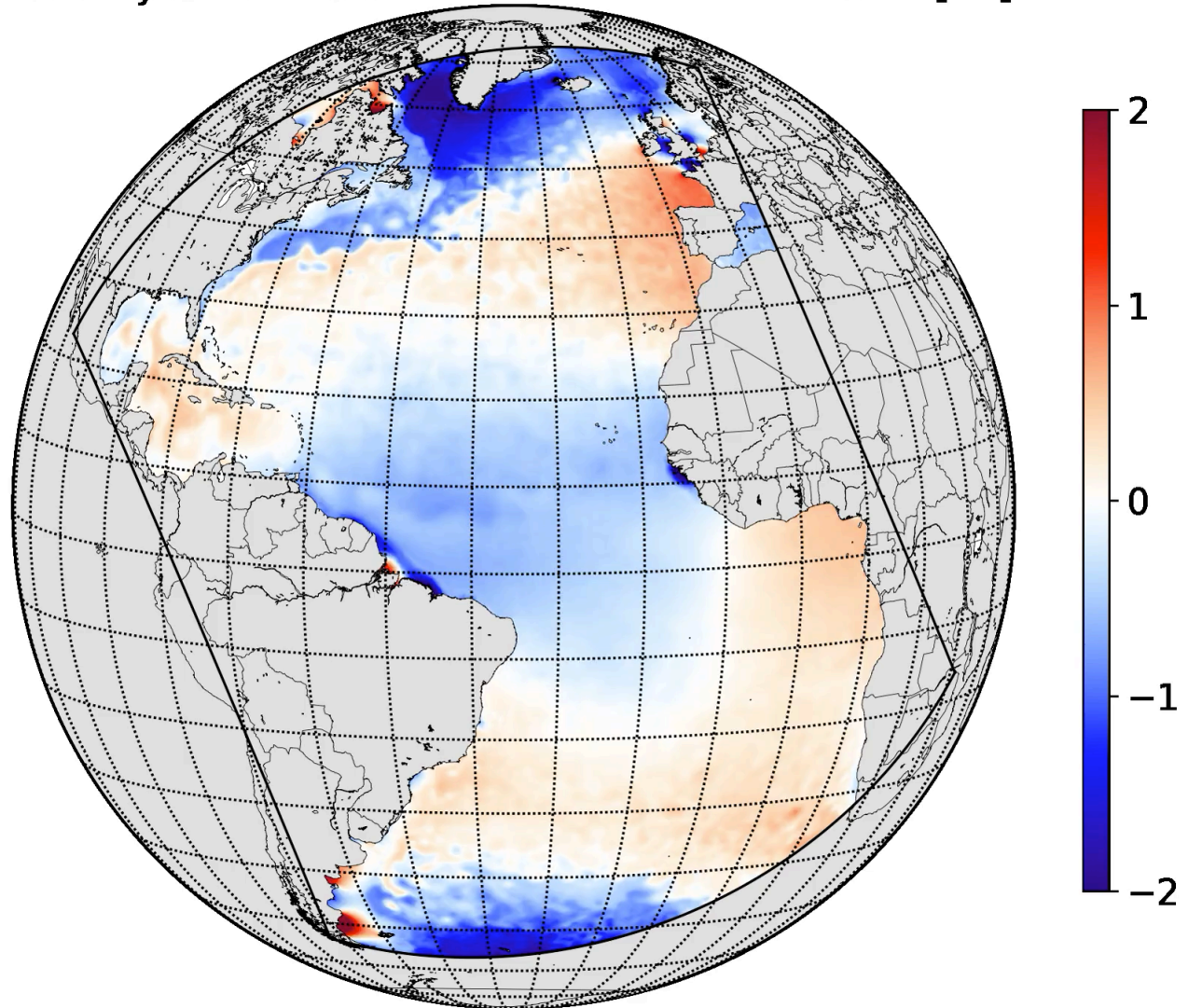


3.4 Coastal waves

Kelvin waves

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Other coastal waves

Wave type	Main restoring mechanism	Cross-shore structure	Propagation
Coastal Kelvin	Gravity + rotation	Exponential decay offshore; no nodes	Coast on right in Northern Hemisphere; left in Southern
Internal Kelvin	Buoyancy + rotation	Exponential decay for each ideal vertical mode	Same direction rule as coastal Kelvin
Continental shelf	Rotation + bottom slope	Trapped over shelf; higher modes can have nodes	Usually the Kelvin-wave direction for a shelf deepening offshore
Stratified coastal-trapped	Buoyancy + rotation + bottom slope	Multiple cross-shore and vertical modes	Depends on mode and background conditions
Beach edge	Gravity over a sloping beach	Trapped near shore; higher modes have nodes	Either alongshore direction